

Unity-MCP API 覆盖矩阵

Pinned: `com.ivanmurzak.unity.mcp` **0.89.0**
Related: [define 矩阵](#) · [示例提示词](#) · [MCP README](#)

Legend: **Direct tool** · **Generic tool** · **Raw only** · **Diagnostic only** · **Out of scope**

MIDI 1.0 (`MidiOutgoingMessageType`)

Enum / API	Coverage	Tool
NoteOn / NoteOff	Direct	<code>midi-send-note</code>
ControlChange	Direct	<code>midi-send-cc</code>
ProgramChange	Direct	<code>midi-send-pc</code>
ChannelAftertouch, PolyphonicAftertouch, PitchWheel	Generic	<code>midi-send-message</code>
SysEx / SystemCommon / SingleByte / MTC / Song* / Tune / Clock / Start/Continue/Stop / ActiveSensing / Reset / Misc / Cable	Generic	<code>midi-send-message</code>
All Sound/Notes Off	Direct	<code>midi-panic</code>
Virtual inject (以上全部)	Generic	<code>midi-virtual-device-inject</code> (+ 便捷工具 <code>midi-virtual-device</code>)
Device name/vendor/product	Direct	<code>midi-device-info</code>
Midi2RawUmp (位于 MIDI1 enum)	Raw only	<code>midi2-send-ump</code>

MIDI 2.0 / UMP (MidiManager send / inject)

Category	Coverage	Tool
Initialize / Terminate / status	Direct	midi2-lifecycle
Device list	Direct	midi2-devices-list
Channel voice (Note/AT/CC/PC/PB/Per-Note/RPN/NRPN relative)	Direct	midi2-send-channel-voice
Utility + MIDI1 System on UMP path	Direct	midi2-send-system
SysEx data	Direct	midi2-send-data
Flex: tempo / time sig / key / text	Direct	midi2-send-flex-data
Flex: metronome / chord name structs	Raw only	midi2-send-ump (结构化 MCP 延后 — 复杂结构体)
Stream: EndpointDiscovery / StartOfClip / EndOfClip	Direct	midi2-send-stream
Stream: Endpoint Info/Identity/Name/Product/FunctionBlock/Config	Raw only	midi2-send-ump
Raw UMP escape hatch	Raw only	midi2-send-ump
Virtual register + inject	Direct	midi2-virtual-device
UMP monitor filter	Direct	midi2-monitor-read

UMP Sequence

API	Coverage	Tool
UmpSequencer play/seek/loop	Direct	ump-sequencer-control
UmpSequencer record/save	Direct	ump-recorder-control
Sequence ↔ UmpSequence	Direct	ump-sequence-convert
Import .midi2 asset	Direct	ump-import-asset

Transport

API	Coverage	Tool
RTP-MIDI start/stop/connect	Direct	<code>midi-rtsp-session</code>
BLE scan/advertise	Direct	<code>midi-ble-control</code> （平台门控）
Nearby discover/advertise	Direct	<code>midi-nearby-control</code>
UDP MIDI 2 / UMP Endpoint	Direct	<code>midi2-udp-session</code> （密钥永不回显）
Aggregate status	Diagnostic	<code>midi-transport-status</code>
<code>FEATURE_ANDROID_COMPANION_DEVICE</code>	Diagnostic only	<code>midi-features-status</code> / docs（构建后处理, 非运行时传输工具）

Foundation / Gameplay

API	Coverage	Tool
MidiDeviceSelection	Direct	<code>midi-device-selection</code>
MidiLatencyCalibrator	Direct	<code>midi-latency-calibration</code>
MidiOutputRoutingPreset	Direct	<code>midi-output-routing</code>
MidiClockSync / MidiClockOutput	Direct	<code>midi-clock-sync</code> / <code>midi-clock-output</code>
ChordRecognition / MidiChordDetector	Direct	<code>midi-chord-state</code>
Note/scale theory	Direct	<code>midi-theory-utility</code> （+ 现有 <code>midi-note-utility</code> ）
SmfPlayer loop/gain/transpose full edit	Out of scope	使用 <code>smf-player-control</code> + Inspector；如有需要可后续添加 <code>smf-player-configure</code>
InputMap binding CRUD（全部消息类型）	Out of scope	现有 <code>midi-setup-router</code> 覆盖 create/append；完整 list/update/remove 延后
Animator mapping list/update/remove	Out of scope	现有 <code>midi-animator-add-*</code> 仍以 append 为导向

Monitor

Item	Coverage	Notes
includeRaw / sinceTimestamp	Direct	<code>midi-monitor-read</code>
MidiMonitorMessageType 扩展（Channel AT / Poly AT / Sys Common/Realtime 拆分）	Out of scope	需要 <code>MidiMonitorMessageType</code> 生产者侧的核心分类变更；Detail 字符串仍为事实来源
Device attach/detach Resource	Out of scope	优先使用 <code>midi-device-info</code> + Play Mode 设备列表

MPTK Free extras

Tool	Coverage
<code>midi-mptk-stream-command</code>	Direct（ <code>MptkUtility</code> ）
<code>midi-mptk-midi2-device</code>	Direct
<code>midi-mptk-mpe-zone</code>	Direct（ <code>ConfigureZone</code> ）
<code>midi-mptk-delayed-dispatch</code>	Direct
<code>midi-mptk-global-settings</code>	Direct（ <code>MPTK_RunInBackground</code> / <code>MPTK_AudioListener</code> ）
<code>midi://mptk/setup-report</code>	ValidateSetup + 诊断（voice stats / SoundFont / global knobs）
<code>midi-mptk-distance-audio</code>	Direct（ <code>MptkDistanceAudioSettings</code> / DistanceAttenuation · Orientation）
<code>midi-mptk-bank-program</code>	Direct（+ <code>action=list-drums</code> 鼓组预设一览）
<code>midi://mptk/soundfont</code>	SoundFont 名 + 鼓组预设一览
<code>midi-mptk-velocity-attenuation</code>	Direct（ experimental <code>MPTK_VelocityAttenuation</code> ；默认 960）

MPTK Pro extras

Tool	Coverage
<code>midi-mptk-effects-configure</code>	Direct（ <code>MptkSynthEffectsController</code> / SoundFont filter·reverb·chorus）
<code>midi-mptk-voice-lifecycle</code>	Direct（ <code>MptkVoiceLifecycle</code> / <code>PauseVoices</code> · <code>ResumeVoices</code> ）
<code>midi-mptk-chord-progression</code>	Direct（ <code>MptkChordProgressionPlayer</code> / Maestro progression presets）

MIDI-CI

Item	Coverage
Discovery	Direct (<code>midi-ci-discover</code>)
Profile / Property / Protocol Negotiation	在库 API 就绪前为 Out of scope

Scriptable Audio / Network (Editor 与 Player)

Tool	Host
<code>midi-sa-bootstrap-*</code>	Editor (接线)
<code>midi-sa-transport</code> / <code>midi-sa-clock-mode</code> / <code>midi-sa-validate</code>	Player 控制 (需已有 Generator)
<code>midi-net-hub-client-setup</code> / <code>midi-net-bridge-setup</code> / <code>midi-wsnet2-sync-defines</code>	Editor (接线 / Define)
<code>midi-net-discovery-status</code> / <code>midi-net-rtt</code>	Player 控制 (<code>ensure*</code> 创建仅 Editor)

VST3

Item	Coverage
Host operations	本仓库 Out of scope

Safety metadata

Tool	Note
<code>midi2-devices-list</code>	已移除 <code>ReadOnlyHint</code> (initialize 有副作用)
<code>mpe-zone-status</code>	已移除 <code>ReadOnlyHint</code> (setUpIfMissing)
<code>midi-net-discovery-status</code>	已移除 <code>ReadOnlyHint</code> (ensure*)
<code>midi-net-rtt</code>	已移除 <code>ReadOnlyHint</code> (ping)
Destructive paths	define 启用、资源覆盖 (UMP convert/save、routing preset) 需 <code>confirm=true</code>